

Sparse-Autoencoder Dissection of Emotion Features in Frozen Speech Encoders

From decodability to mechanism

Anonymous

Frozen speech encoders (HuBERT, WavLM, wav2vec2, w2v-BERT, MERT, Whisper) top SER benchmarks — but they were trained for **ASR**, so they encode lots of *text*.

- Probing tells us **where** emotion is readable. It cannot tell us:
- *Which* features encode emotion?
- Is the signal **acoustic** (how it's said) or **lexical** (what's said)?
- Does the pattern hold across encoders and speakers?

Idea: use *Sparse Autoencoders* to go from **decodability** → **mechanism**.

- Train a **TopK SAE** ($8\times$, $k=32$) on **per-frame** activations of each frozen encoder ($\sim 940k$ frames). FVU 0.07 (93% variance), 4/8192 dead.
- Anchor on **CREMA-D**: 91 actors $\times$ 12 *fixed* sentences $\times$ 6 emotions $\Rightarrow$ text is held constant, so any emotion signal is *acoustic* (built-in control).
- **Confound-controlled emotion features**: keep a feature only if its firing tells us more about *emotion* than about the *sentence* (phonetics) or the *speaker*.

394 naive features

↓ confound control

61 genuine emotion features
(median monosemanticity 0.59)

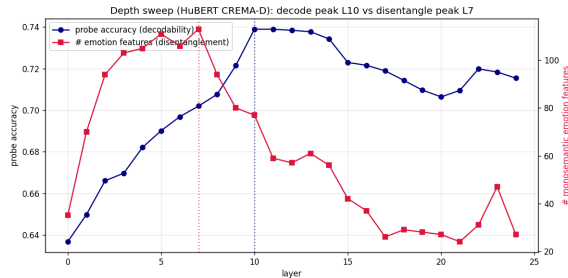
dominated by **anger** & **sadness**;
fear/neutral $\rightarrow$ 0

Without confound control, $\sim 66\%$ of “emotion” features are really phonetic/speaker features — the classic “SAE rediscovers phonemes, not affect” trap.

Q1 — Depth: decoding $\neq$ disentangling

- Probe accuracy peaks **deep** (L10) and stays high.
- But clean monosemantic emotion features peak **early** (L7; monosemanticity L2).
- Deep layers keep *decodability* but lose *disentanglement* (24–47 feats).

Takeaway: emotion is *organised* into clean features earlier than where a probe reads it best.

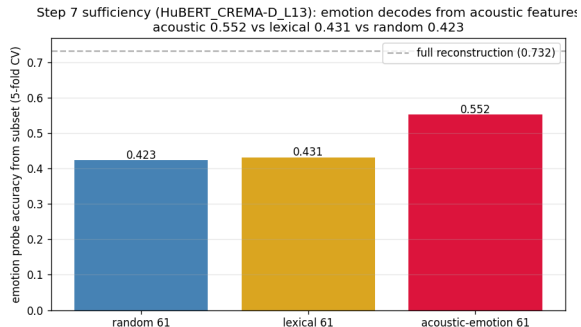


- All six encoders carry emotion features (61–145 each), all anger/sadness-dominated.
- Cross-encoder feature overlap: **Whisper**→**SSL 0.40** $\approx$ **SSL**→**SSL 0.39** — on fixed-lexicon speech Whisper is *not* special; its difference is about *text*, not acoustics.
- **Speaker-invariant:** leave-speaker-out (91 actors) accuracy drop ≈ 0 ; each feature fires for $\sim 95\%$ of its emotion's actors.
- $\Rightarrow$ the published MESD leave-one-speaker-out collapse (~ 0.30) is due to the *speaker-entangled* features our control removes.

Q3 — The signal is acoustic, not lexical

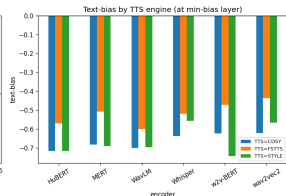
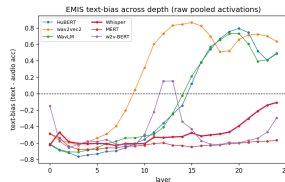
- Positive control: **60/61** features label *acoustic* (eGeMAPS AUC 0.83 vs sentence 0.66).
- Causal *sufficiency*: emotion decodes at **0.55** from acoustic features vs **0.43** lexical vs **0.42** random (full 0.73).

Takeaway: what the probe exploits is paralinguistic delivery, not words.



Q4 — EMIS: the lexical shortcut is a deep-layer thing

- EMIS = synthetic speech where *text* emotion $\neq$ *voice* emotion.
- Train on congruent, test on incongruent; text-bias = text–audio acc.
- Shortcut appears **only in deep layers** of ASR-SSL (HuBERT/WavLM/wav2vec2, +0.7–0.9).
- **Whisper stays audio-grounded at every layer** (L24 –0.11); MERT/w2v-BERT never take it.
- Reproduces the published text-bias — and *localizes* it to depth.



- **AudioSAE** (EACL 2026): same SAE recipe on 2 encoders, but emotion is just one probe task — *no* emotion-feature interpretation, confound control, factorial corpus, or acoustic/lexical split. Closest method neighbor.
- ASR-SAE works (Pluth'26; Mariotte'26): SAEs on speech, but find phonetic/lexical or singing features — no affect, no depth sweep.
- Ma et al. '25: mechanistic Whisper-SER — but via probing/CKA, **no SAEs**.

Un-scooped core: confound-controlled SAE emotion features on a lexicon-controlled factorial corpus across 6 encoders + $\text{decode} \neq \text{disentangle}$ depth + EMIS-shortcut-by-depth.

Limitations

- Necessity-ablation is near-null (signal distributed) — we rely on sufficiency.
- SAE reconstruction must *not* replace raw activations for the EMIS probe (it inverted Whisper).
- Cross-encoder/EMIS arms use one layer per encoder, not full 25-layer grids.

Future work

- IEMOCAP real-transcript causal ablation (stub ready; needs transcripts).
- SpeechCraft cross-lingual feature overlap → explain EN↔ZH collapse.
- Feature *steering*; dimensional V/A/D targets; full per-encoder layer grids.

- ① SAEs move SER analysis from *decodability* to *mechanism*; confound control is essential.
- ② **Decoding $\neq$ disentangling**: emotion organises early (L2–L7), decodes best deep (L10).
- ③ Emotion features are **acoustic** and **speaker-invariant**, shared across all six encoders.
- ④ The **lexical shortcut** is a deep-layer ASR-SSL phenomenon; **Whisper resists it everywhere**.

All code (`src/sae/`), data pointers, artifacts (`results/SAE/`) and the report are released on the SAE branch.